

Real-World Real Estate

PILOT PLAN



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Purpose and success definition

This Pilot Plan defines a milestone-aligned approach to validate the platform end-to-end under real project constraints.

Pilot objectives:

- Milestone workflows reflect real construction and approval practices.
Evidence intake and proof generation are consistent, repeatable, and reproducible.
- Independent verification is possible using the recorded fingerprint and on-chain reference (TXID).
- Certification rules are enforceable in practice (validation-critical milestones require complete evidence and appropriate attestations).
- Outputs are usable for downstream professional workflows.
- Exception handling is transparent and auditable (Observed/remediation loops, delays, supersession).

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Roles and responsibilities

Clear role definitions keep actions attributable, access properly scoped, and evidence traceable end to end—from submission and certification through verification, and audit logging.

Real Estate Developer Partner 1 and Partner 2

Own the project-side inputs: define the project context (units/milestones as configured), supply and upload evidence over time, ensure validation-critical artifacts are uploaded only once they exist in certified/stamped/signed form by the competent authority or professional, respond to Observed items with remediation evidence, and confirm the resulting export package for their unit is usable for professional review.

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Council of Experts

Provides lightweight governance through the three core signatories: Developer 1, Developer 2, and the Notary, reviewing stage-gate deliverables for practical fit with development and notarial/registrar expectations.

Platform Operator / Admin

Day-to-day operation: onboards participants, provisions accounts, enforces role- and unit-scoped access, operates off-chain services (storage, evidence intake, proof generation, audit logging), coordinates the schedule and evidence capture, and ensures exports and logs are produced when required.

3 Purpose and success definition

This section defines the product capabilities that must exist so the workflow can be executed end to end.

3.1 Access control and account model

The platform provides identity-bound user accounts and a permission model that is enforced consistently across units, milestones, evidence, exports, and logs.

- A user's visibility is scoped to the unit or units they are assigned to, and further constrained by role permissions.
- Roles govern which actions a user can perform, such as uploading evidence, changing milestone state, sharing access, exporting a dossier, and reviewing audit history.

- Access changes are treated as first-class events: invitations, access grants, and access revocations are recorded as part of the system's traceable history.

3.2 Core objects and data relationships

The MVP supports a domain model that can express a real development in a way that is reviewable and verifiable.

- A development contains one or more units.
- Each unit has a timeline composed of milestones.
- Each milestone has a state and may have required evidence categories.
- Evidence is an object with metadata and links to one or more milestones.
- Where bundling is used, the system can represent a bundle as a defined set of evidence items plus a deterministic commitment over that set.
- A unit can produce a dossier export that includes references back to milestones, evidence, attestations, and anchoring references.

3.3 Milestone workflow and exception handling

The MVP implements a constrained state model for milestones, with a time-ordered history.

- Milestones move through a defined set of states used by the workflow, including a path for issues and remediation.
- State changes are attributable to an actor and recorded as a sequence, so the unit timeline can be reconstructed and reviewed later.
- Exception handling is represented explicitly through an Observed path that captures the issue, links remediation evidence, and preserves the full ordering of events.

3.4 Evidence management, taxonomy, and linkage

Evidence handling is structured so the platform can serve as a traceability layer.

- Evidence is categorized using a consistent taxonomy so reviewers can understand what kind of artifact they are looking at and why it matters.
- Evidence is linked to milestones rather than only to the unit, so the timeline can be navigated milestone by milestone and each milestone can surface its supporting artifacts.
- Evidence supports practical review needs such as supersession: newer versions can be added while older versions remain available for traceability and dispute resolution.
- Evidence visibility follows access control rules, so a buyer or investor can view what has been shared for their unit without gaining visibility into unrelated unit data.

3.5 Integrity proofs and anchoring references

The MVP generates integrity markers for evidence and records on-chain anchoring references so that later verification is possible.

- The system computes deterministic fingerprints for evidence artifacts. Where bundling is used, it computes a deterministic commitment for the bundle.
- It records the anchoring reference associated with each fingerprint or commitment and links that reference back to the relevant milestone or evidence context.
- On-chain data is restricted to integrity and reference information; documents and sensitive content remain off-chain under access control.

3.6 Independent verification workflow

The MVP supports a reviewer path that can validate integrity without relying on trust in the operator.

- A reviewer can obtain the permitted evidence through the platform's authorized access path.
- The reviewer can recompute the fingerprint or verify inclusion under a bundle commitment using the exported materials.
- The reviewer can compare the recomputed values to what the platform records as the anchoring reference for that evidence or bundle.
- The system provides enough context in its records and exports for a reviewer to perform this comparison reliably.

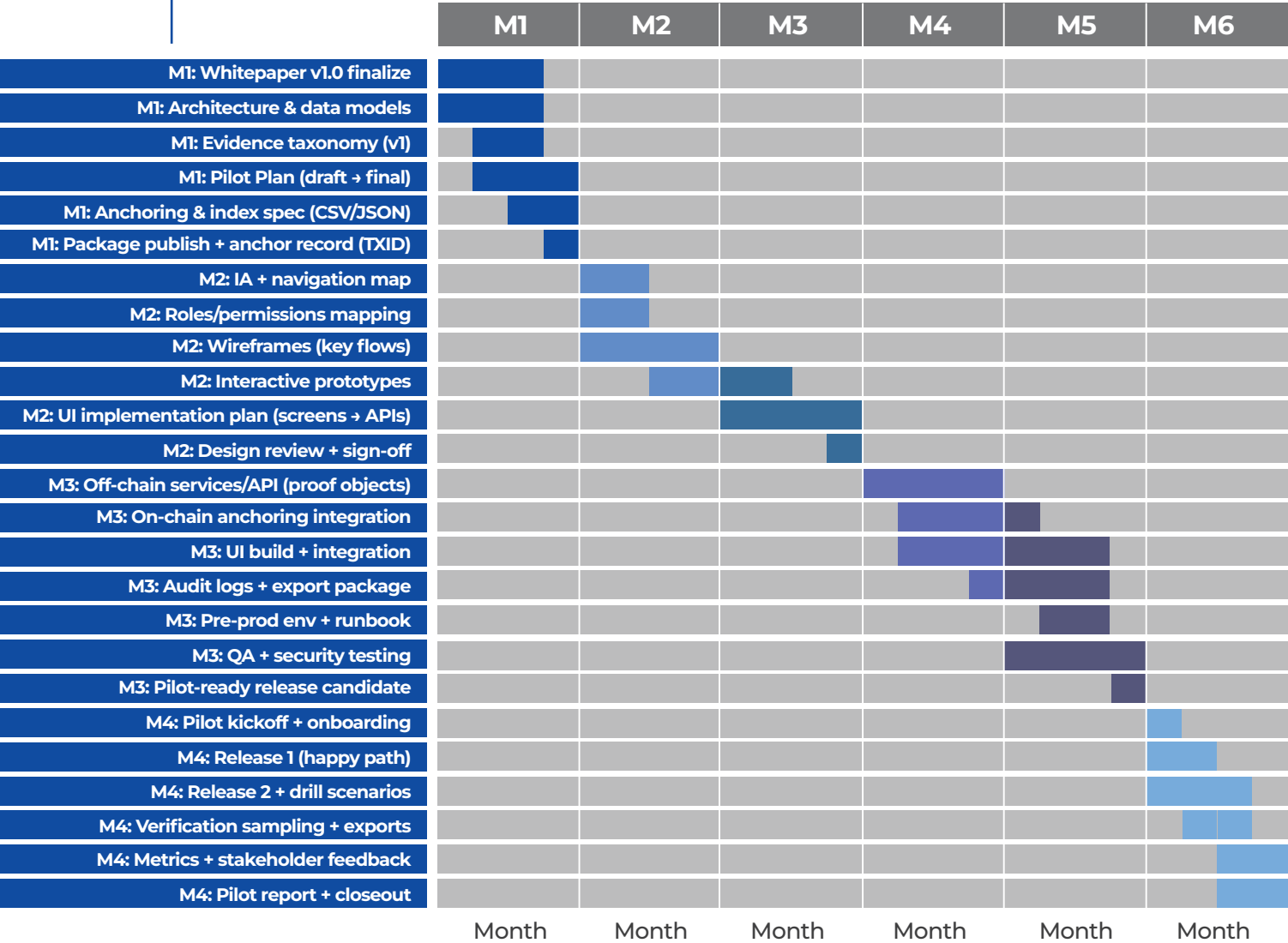
3.7 Dossier export package

The MVP can generate a dossier package for a unit intended for professional review.

- The dossier includes a milestone narrative or index, evidence manifests, and references to attestations and anchoring references.
- It is designed to be coherent when consumed outside the platform, so a reviewer can follow the chain from milestone to evidence to integrity reference.

4 Timeline (Gantt) and milestone execution plan

Schedule structure (fixed durations): Milestone 1 = 1 month; Milestone 2 = 2 months; Milestone 3 = 2 months; Milestone 4 = 1 month.



Milestone breakdown (activities, outputs, and

Milestone 1 (Month 1) - Definition package

Key activities:

- Finalize Whitepaper v1.0 (pilot contract, workflow, policies).
- Architecture and data models (needed for build and verification).
- Evidence taxonomy v1 (categories + required evidence checklist structure).
- Pilot Plan v1.1 (this document).
- Anchoring and public-location rules; CSV/JSON index specification.
- Publish M1 artifacts and record anchors (TXIDs) in the deliverables index.

Gate: M1 package is published and the deliverables index format is confirmed (finger-prints and TXIDs recorded).

Milestone 2 (Months 2-3) - UX and Product Design System

Key activities:

- ☐ Information architecture (IA) and navigation map for core flows.
- ☐ Role/permission mapping aligned to the pilot roles.
- ☐ Wireframes for key flows (onboarding, unit timeline, evidence upload, certification gating, exports).
- ☐ Interactive prototypes used as acceptance references.
- ☐ UI implementation plan mapping screens/components to API endpoints and test cases.

Gate: Prototypes and implementation plan are complete and can be used as the reference for build and testing.

Milestone 3 (Months 4-5) - Build, integration, and pre-production readiness

Key activities:

- ☐ Implement off-chain services/APIs for fingerprinting and proof objects (pilot-ready).
- ☐ Integrate on-chain anchoring; ensure TXID recording and retrievability.
- ☐ Implement UI and integrate with services (core flows only).
- ☐ Implement audit logs and export package generation.
- ☐ Set up pre-production environment and runbook.
- ☐ QA and security testing sufficient for controlled pilot execution; produce release candidate.

Gate: Pre-production build passes the acceptance test suite and is ready for controlled pilot execution.

Milestone 4 (Month 6) - Pilot execution and closeout package

Key activities:

- ☐ Pilot kickoff and onboarding of real stakeholders and real project data (as defined in the pilot setup).
- ☐ At least two iterative releases during the pilot window (happy path + drill scenarios).
- ☐ Run drill scenarios (re-inspection, supersession, delayed approvals, partial evidence, remediation loops).

- Verification sampling and dossier/export generation for pilot units.
- Collect metrics and stakeholder feedback; produce pilot report and closeout package; update documentation as needed.

Gate: Pilot outputs are produced, verifiability is demonstrated, and the closeout package is complete.

5 Metrics and measurement plan

Metrics are defined to measure pilot performance and verifiability and to support clear, audit-ready reporting. Where a metric is expressed as a rate or time, it is reported as an observed distribution and reviewed alongside the acceptance test outcomes

VERIFICATION SUCCESS RATE

Definition: Percent of sampled evidence bundles whose recomputed fingerprint matches the recorded fingerprint and anchoring reference (TXID).

Source: Verification log + deliverables index (CSV/JSON).

Frequency: Per release + end of pilot

Success interpretation: No unexplained mismatches; mismatches are detected, logged, and resolved with notes.

OBSERVED COUNT AND RESOLUTION TIME

Definition: Number of Observed events and time from Observed to documented outcome (Certified or documented resolution).

Source: Audit log / state transitions.

Frequency: Weekly From pilot start + end of pilot

Success interpretation: All Observed items have a recorded outcome; resolution time is reported.

EVIDENCE COMPLETENESS RATE

Definition: Percent of certification attempts where the required evidence set is complete at the time of certification attempt.

Source: Milestone checklist records.

Frequency: Per release + end of pilot

Success interpretation: Incomplete certification attempts are blocked; completeness is reported.

STAKEHOLDER FEEDBACK

Definition: Structured feedback from pilot participants (clarity, usability, trust, review readiness).

Source: Post-pilot survey/interviews.

Frequency: End of pilot

Success interpretation: Feedback collected and summarized with actionable issues.

6 Acceptance test cases (pass/fail)

TC-RBAC-01 — Unit-bound access

The goal here is to prove Role-Based Access Control (who can see/do what) is enforced at the unit level: when a buyer/investor is invited to one specific unit, they should only be able to view that unit’s timeline and evidence, and nothing outside of it.

Area: Access control

Steps (high level): Invite buyer/investor to a specific unit; login; attempt to access other unit data.

PASS criteria: User can view only the assigned unit timeline/evidence; unauthorized access is denied and logged.

FAIL criteria: User can access non-assigned unit data OR denial is not logged.

Evidence captured: Screenshots of access boundaries; audit log entry.

TC-WF-01 — Milestone state progression

This scenario validates the milestone Workflow by walking through the standard lifecycle and confirming that state changes follow the allowed order, with a record that clearly shows who performed each transition and when.

Area: Workflow

Steps (high level): Create milestone; move Pending → In Progress → Certified following checklist.

PASS criteria: State transitions succeed only in valid order and are time-stamped with actor attribution.

FAIL criteria: Invalid transition is allowed OR actor/timestamp is missing from the record.

Evidence captured: State history export; audit log snippet.

TC-OBS-01 — Observed and remediation loop

Here the focus is the **Observed** path: a milestone is flagged with a reason, remediation evidence is added, and the milestone returns to progress for re-certification—without breaking the audit trail or losing the time-ordered story of what happened.

Area: Exceptions

Steps (high level): Mark milestone Observed with reason; add remediation evidence; return to In Progress and re-attempt certification.

PASS criteria: Observed event is recorded; remediation is linked; outcome is auditable and time-ordered.

FAIL criteria: Observed event is not recorded OR remediation/outcome cannot be audited.

Evidence captured: Observed record + resolution record; evidence links.

TC-EVID-01 — Evidence upload and linkage

This test is about **Evidence** behaving correctly as an auditable object: uploads must be stored reliably, linked to the right milestone, visible only to authorized viewers under sharing rules, and accompanied by logged sharing events.

Area: Evidence

Steps (high level): Upload evidence items; link them to a milestone; verify metadata and visibility under sharing rules.

PASS criteria: Evidence is stored and linked; correct viewers can access; sharing events are logged.

FAIL criteria: Evidence cannot be retrieved OR linkage is incorrect OR sharing is not logged.

Evidence captured: Evidence list export; sharing log entry.

TC-PROOF-01 — Fingerprint and TXID recorded

The emphasis here is the **Proof** trail: an evidence bundle produces a stable fingerprint (for example, SHA-256), an anchor is created, and the TXID reference is recorded in a way that can later be retrieved from the milestone/evidence view. The “01” identifier tracks this as the first proof-generation scenario.

Area: Anchoring

Steps (high level): Create evidence bundle; generate fingerprint (e.g., SHA-256) and anchor; store TXID reference.

PASS criteria: Fingerprint is deterministic; TXID reference is recorded and retrievable from the milestone/evidence view.

FAIL criteria: Fingerprint differs for identical bundle OR TXID is missing/unretrievable.

Evidence captured: Deliverables index entry; milestone screen screenshot.

TC-AUD-01 — Audit trail for key actions

This scenario is about **Auditability**: after performing core actions (share, upload, state change, export), the audit trail should show actor, timestamp, action type, and the object reference for each event, consistently. “01” tracks it as the first audit-focused scenario.

Area: Auditability

Steps (high level): Perform share, upload, state change, and export actions; review audit trail.

PASS criteria: Audit trail records actor, timestamp, action, and object reference for each action.

FAIL criteria: Audit trail missing actor/timestamp/object reference for any key action.

Evidence captured: Audit log export.